

Module3_Lab1

September 17, 2026

```
[1]: import pandas as pd
df = pd.read_csv("messy_sales.csv")
df.head()
```

```
[1]:   order_id      date  product  price  qty  zip
0      1254  04/06/2026      mug  11.36   4  60614
1      1057  2026-05-30  charger  14.79   3  30303
2      1150  05/06/2026      mug   5.50   1  10001
3      1066  05/30/2026  notebook   NaN   1  98101
4      1114  04/06/2026  webcam  45.59   4  90405
```

```
[2]: import pandas as pd
df = pd.read_csv("messy_sales.csv")
df.shape
```

```
[2]: (300, 6)
```

```
[3]: import pandas as pd
df = pd.read_csv("messy_sales.csv")
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 300 entries, 0 to 299
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   order_id    300 non-null    int64
1   date        300 non-null    str
2   product     300 non-null    str
3   price       288 non-null    float64
4   qty         300 non-null    int64
5   zip         300 non-null    int64
dtypes: float64(1), int64(3), str(2)
memory usage: 19.2 KB
```

```
[4]: import pandas as pd
df = pd.read_csv("messy_sales.csv")
df.describe()
```

```
[4]:
```

	order_id	price	qty	zip
count	300.000000	288.000000	300.000000	300.000000
mean	1144.816667	56.068438	2.913333	45385.276667
std	84.515319	72.307182	1.560107	39345.608641
min	1000.000000	2.930000	-5.000000	2116.000000
25%	1071.750000	10.945000	2.000000	2134.000000
50%	1144.500000	37.530000	3.000000	30303.000000
75%	1217.250000	61.417500	4.000000	90210.000000
max	1291.000000	479.290000	5.000000	98101.000000

After running the first commands on this new csv dataset, I can tell that there are 300 rows, 6 columns worth of sales data. It provides us the Order ID, Date, Product Price, Qty, and Zip. For instance, there are a number of missing/blank slots for price, there are some negative values under Quantity, which is impossible, and under Zip some are 4 digits and most 5 digits.

```
[5]: df.isnull().sum()
```

```
[5]: order_id    0
      date       0
      product    0
      price     12
      qty        0
      zip        0
      dtype: int64
```

```
[6]: df.isnull().sum()
```

```
[6]: order_id    0
      date       0
      product    0
      price     12
      qty        0
      zip        0
      dtype: int64
```

```
[7]: df['price'] = df['price'].ffill()
```

```
[8]: df.isnull().sum()
```

```
[8]: order_id    0
      date       0
      product    0
      price      0
      qty        0
      zip        0
      dtype: int64
```

I decided to use the forward fill function for this. I thought about putting zero since we don't know. However, that might heavily skew my data.

```
[9]: print(df.duplicated().sum())
df = df.drop_duplicates()
```

8

```
[10]: df.shape()
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[10], line 1
----> 1 df.shape()

TypeError: 'tuple' object is not callable
```

```
[11]: df.shape
```

```
[11]: (292, 6)
```

```
[12]: df["zip"] = df["zip"].astype(str).str.zfill(5)
```

```
[13]: df["date"] = pd.to_datetime(df["date"], format="mixed")
```

```
[14]: df.head()
```

```
[14]:
```

	order_id	date	product	price	qty	zip
0	1254	2026-04-06	mug	11.36	4	60614
1	1057	2026-05-30	charger	14.79	3	30303
2	1150	2026-05-06	mug	5.50	1	10001
3	1066	2026-05-30	notebook	5.50	1	98101
4	1114	2026-04-06	webcam	45.59	4	90405

```
[15]: df[df["qty"] < 0]
```

```
[15]:
```

	order_id	date	product	price	qty	zip
202	1140	2026-04-18	webcam	38.69	-5	98101
262	1233	2026-05-09	desk lamp	22.64	-4	10001
297	1025	2026-04-27	keyboard	47.30	-2	2116

```
[16]: df['qty'] = df['qty'].abs()
```

```
[17]: df[df["qty"] < 0]
```

```
[17]: Empty DataFrame
```

```
Columns: [order_id, date, product, price, qty, zip]
```

```
Index: []
```

300 rows of data with 6 columns related to sales of notebooks, webcams, monitors, pen sets, etc. were cleaned, using forward fill to fill missing price data, set the zip codes to the correct 5 digits,

formatted the correct dates to standard format. Negative values were corrected. Lastly, I decided to use the forward fill function for this. Making the wrong decision could have been a disaster to the business, for instance, I thought about putting zero since we don't know. However, that might heavily skew my data.